

Breaking change: BinaryFormatter serialization methods are obsolete and prohibited in ASP.NET apps

Breaking change: BinaryFormatter serialization methods are obsolete and prohibited in ASP.NET apps - gewarren, learn.microsoft.com.

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Content

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BinaryFormatter serialization methods are obsolete and prohibited in ASP.NET apps

Summarize this article for me

`Serialize` and `Deserialize` methods on [BinaryFormatter](#), [Formatter](#), and [IFormatter](#) are now obsolete as warning. Additionally, [BinaryFormatter](#) serialization is prohibited by default for ASP.NET apps.

Note

In .NET 7, the [affected APIs]() are obsolete as *error*. For more information, see [BinaryFormatter serialization APIs produce compiler errors](#).

Change description

Due to [security vulnerabilities](#) in [BinaryFormatter](#), the following methods are now obsolete and produce a compile-time warning with ID `SYSLIB0011`. Additionally, in ASP.NET Core 5.0 and later apps, they will throw a [NotSupportedException](#), unless the web app has re-enabled [BinaryFormatter](#) functionality.

- [BinaryFormatter.Serialize](#)
- [BinaryFormatter.Deserialize](#)

The following serialization methods are also obsolete and produce warning `SYSLIB0011` , but have no behavioral changes:

- [Formatter.Serialize\(Stream, Object\)](#)
- [Formatter.Deserialize\(Stream\)](#)
- [IFormatter.Serialize\(Stream, Object\)](#)
- [IFormatter.Deserialize\(Stream\)](#)

Version introduced

5.0

Reason for change

These methods are marked obsolete as part of an effort to wind down usage of [BinaryFormatter](#) within the .NET ecosystem.

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Stop using [BinaryFormatter](#) in your code. Instead, consider using [JsonSerializer](#) or [XmlSerializer](#). For more information, see [BinaryFormatter security guide](#).

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You can temporarily suppress the [BinaryFormatter](#) compile-time warning, which is `SYSLIB0011` . We recommend that you thoroughly assess your code for risks before choosing this option. The easiest way to suppress the warnings is to surround the individual call site with `#pragma` directives.

```
// Now read the purchase order back from disk
using (var readStream = new FileStream("myfile.bin", FileMode.Open))
{
    var formatter = new BinaryFormatter();
    #pragma warning disable SYSLIB0011
    return (PurchaseOrder)formatter.Deserialize(readStream);
    #pragma warning restore SYSLIB0011
}
```

You can also suppress the warning in the project file.

```
<PropertyGroup>
  <OutputType>Exe</OutputType>
  <TargetFramework>net5.0</TargetFramework>
  <!-- Disable "BinaryFormatter is obsolete" warnings for entire project -->
  <NoWarn>$(NoWarn);SYSLIB0011</NoWarn>
</PropertyGroup>
```

If you suppress the warning in the project file, the warning is suppressed for all code files in the project. Suppressing `SYSLIB0011` does not suppress warnings caused by using other obsolete APIs.

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To continue using [BinaryFormatter](#) in ASP.NET apps, you can re-enable it in the project file. However, it's strongly recommended not to do this. For more information, see [BinaryFormatter security guide](#).

```
<PropertyGroup>
  <TargetFramework>net5.0</TargetFramework>
  <!-- Warning: Setting the following switch is *NOT* recommended in web apps. -->

  <EnableUnsafeBinaryFormatterSerialization>true</EnableUnsafeBinaryFormatterSerialization>
</PropertyGroup>
```

For more information about recommended actions, see [Resolving BinaryFormatter obsoletion and disablement errors](#).

Affected APIs

- [System.Runtime.Serialization.Formatters.Binary.BinaryFormatter.Serialize](#)
- [System.Runtime.Serialization.Formatters.Binary.BinaryFormatter.Deserialize](#)
- [System.Runtime.Serialization.Formatter.Serialize\(Stream, Object\)](#)
- [System.Runtime.Serialization.Formatter.Deserialize\(Stream\)](#)
- [System.Runtime.Serialization.IFormatter.Serialize\(Stream, Object\)](#)
- [System.Runtime.Serialization.IFormatter.Deserialize\(Stream\)](#)

See also

- [SerializationFormat.Binary is obsolete \(.NET 7\)](#)
- [BinaryFormatter serialization APIs produce compiler errors \(.NET 7\)](#)
- [BinaryFormatter disabled across most project types \(.NET 8\)](#)

